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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VII Semester B. E. Examinations Feb/Mar - 2023
Computer Science and Engineering

PARALLEL ARCHITECTURE AND DISTRIBUTED PROGRAMMING

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Define feature size.	01
	1.2	Assume a 20cm diameter wafer has a cost of 15, contains 100 dies and has 0.031 defects/cm ² . Find the yield of the wafers.	02
	1.3	For the following code snippet, determine which accesses are likely to cause data cache misses. for(i=0;i<3;i=i+1) for(j=0;j<100;j=j+1) a[i][j]=b[j][0]*b[j+1][0];	02
	1.4	Differentiate between temporal locality and spatial locality.	02
	1.5	Define Trace Scheduling.	01
	1.6	List the orderings that sequential consistency needs to maintain.	02
	1.7	The number and size of tasks into which a problem is decomposed determines the _____ of the decomposition.	01
	1.8	What are required MPI routines to generate a minimum parallel code?	02
	1.9	Describe the purpose of MPI_BCAST.	01
	1.10	Differentiate between UMA and NUMA.	02
	1.11	State owner computes rule.	01
	1.12	In OpenCL, the kernel is launched with a two-dimensional _____ containing all work items.	01
	1.13	The OpenCL kernel calls the API functions _____ & _____ to fetch the total number of work items in the x-and y-dimension of all work items.	02

PART-B

2	a	Assume a disk subsystem with the following components and MTTF: ▪ 10 disks, each rated at 1,000,000-hour MTTF. ▪ 1 ATA controller, 5000,000-hour MTTF. ▪ 1 power supply, 200,000-hour MTTF ▪ 1 fan, 200,000-hour MTTF. ▪ 1 ATA cable, 1,000,000-hour MTTF. Using the simplifying assumptions that the lifetimes are exponentially distributed and that failures are independent, compute the MTTF of the system as a whole.	04
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	b	With a neat labeled diagram explain the basic structure of a floating point unit using Tomasulo's algorithm extended to handle speculation.	08
	c	How many bits are in the (0,2) branch predictor with 4K entries? How many entries are in a (2,2) predictor with the same number of bits?	04
3	a	Inorder to achieve a speedup of 80 with 100 processors, determine what fraction of the original computation can be sequential?	05
	b	List the key ideas involved in relaxed consistency model. Describe four set of ordering by which the relaxed consistency model relaxes.	06
	c	Differentiate between coherence and consistency. Discuss under what scenario's would a memory system be called coherent?	05
		OR	
4	a	With a neat state transition diagram, explain the individual cache block in a directory based system.	08
	b	Discuss briefly the hardware primitives for implementing synchronization in multiprocessors. Give the code sequence that implements an atomic exchange by employing a pair of these special instructions (an atomic exchange on the memory location).	08
5	a	With a neat diagram explain the process of scheduling threads of SIMD instructions.	08
	b	Draw the basic structure of vector architecture for VMIPS and explain its components.	08
		OR	
6	a	Justify the need of scalar registers included as a primary component in VMIPS architecture. Discussing the term 'code is vectorized', provide the VMIPS code considering the code segment given below. Assume that the starting addresses of m, s, t and u are in Rm, Rs, Rt and Ru, respectively and MVL is 64. <pre> for (j = 0; j < 512; ++j) { x[i] = m[i] * s[i] - t[i] * u[i]; y[i] = m[i] * u[i] + t[i] * s[i]; } </pre>	10
	b	Discuss the major innovations introduced by Fermi to GPU architecture.	06
7	a	Discuss the appropriate examples, the various forms of decomposition based on data or that operate on large data structures.	10
	b	Describe how MPI splits a group of processes in a communicator into subgroups.	06
8	a	With a neat figure discuss OpenCL parallel execution model.	07
	b	With suitable diagram of conceptual OpenCL device architecture, explain how OpenCL models a heterogeneous parallel computing system as a host and one or more OpenCL devices.	09